

Prashant

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EDUCATION

- Vellore Institute of Technology** Bhopal, India
Bachelor of Technology - Computer Science and Engineering; CGPA: 8.67 Sep 2023 - May 2027
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Databases, Computer Networks
- Modern Vidya Niketan School** Faridabad
Class XII (CBSE) July 2023
- Vidya Niketan School** Faridabad
Class X (CBSE) July 2021

SKILLS

- Programming:** Python(Basics), C++ , OOPS ,JavaScript, Problem Solving
- Backend:** Node.js, Express.js, MongoDB, REST APIs, Authentication & Authorization
- Data Structures and Concepts:** Data Structures, Algorithms, Object Oriented Programming, Database Management System, Operating System, Computer Networks
- Databases:** MySQL / SQL, Basic query optimization, joins
- Modeling:** UML, ER Diagrams, Data Flow Diagrams
- DevOps & Tools:** Git, AWS(Decent), Postman, MS-SQL, Firebase,Excel,Notion,Draw.io,Jira(Basics)
- Concepts:** SDLC, API Integration, System Design Basics,Requirement Analysis

PROJECTS

- DPI — Deep Packet Inspection System** **GitHub:** Developed a high-performance Deep Packet Inspection (DPI) engine to analyze and filter network traffic from PCAP files using C++. Designed a robust packet processing pipeline to parse Ethernet, IP, and TCP/UDP protocols, enabling deep inspection of packet-level data.
Implemented flow-based tracking using the Five-Tuple model to manage stateful connections and ensure accurate tracking of network sessions. Integrated TLS SNI extraction to identify applications such as YouTube and Facebook from encrypted HTTPS traffic, enabling intelligent traffic classification.
Engineered a rule-based filtering system to block traffic based on IP, domain, and application type, supporting fine-grained network control. Designed and implemented a multi-threaded architecture using load balancers and fast-path worker threads to process packets in parallel.
Utilized thread-safe queues, mutexes, and condition variables to ensure safe and efficient inter-thread communication. Generated detailed traffic analytics reports including protocol distribution, application breakdown, and packet statistics.
Tech: C++, Multithreading, TCP/IP, TLS, PCAP, Data Structures, Concurrency, System Design **Sep 2025**
- WaveTune — Modern Interactive Music Player** **GitHub:** Developed a stylish and interactive music player to enhance user experience with visually engaging elements and intuitive controls. Designed a responsive UI using HTML and CSS, ensuring seamless interaction across devices.
Implemented dynamic playback features using JavaScript, including play, pause, skip, and volume control. Optimized smooth animations and transitions to deliver a modern and visually appealing audio experience.
Tech: HTML, CSS, JavaScript **Apr 2026**

CERTIFICATIONS

- Market Analysis – NPTE(IIT)** May 2026
- Google IT Support Professional Certificate – Google (Google Skills)** Jan 2026
- The Bits and Bytes of Computer Networking – Google (Coursera)** Dec 2025
- Introduction to Machine Learning – NPTEL (IIT)** Sep 2025
- AWS Technical Essentials – AWS Educate (Amazon)** June 2025
- Operating Systems and Systems Programming – Cursa** Apr 2025
- Frontend Development Libraries – freeCodeCamp** Sep 2024

POSITIONS OF RESPONSIBILITY

- Cultural Representative** April 2025 – Present
Pahadi Club, VIT Bhopal, India
- Led cultural events, fostering teamwork, collaboration, and cross-functional coordination.
 - Strengthened project management, leadership, and communication skills through planning and executing campus initiatives.